



AIAA3201

Introduction to Computer Vision

Week 1

Hao WANG
AI Thrust, INFO Hub



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- TA
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DO Bring Your
Own Laptop!

WeChat Group



群聊：AIAA 3201



该二维码7天内(9月7日前)有效，重新进入将更新

Today's Journey



Cool Applications



History of CV



Course Plan



Image Basics

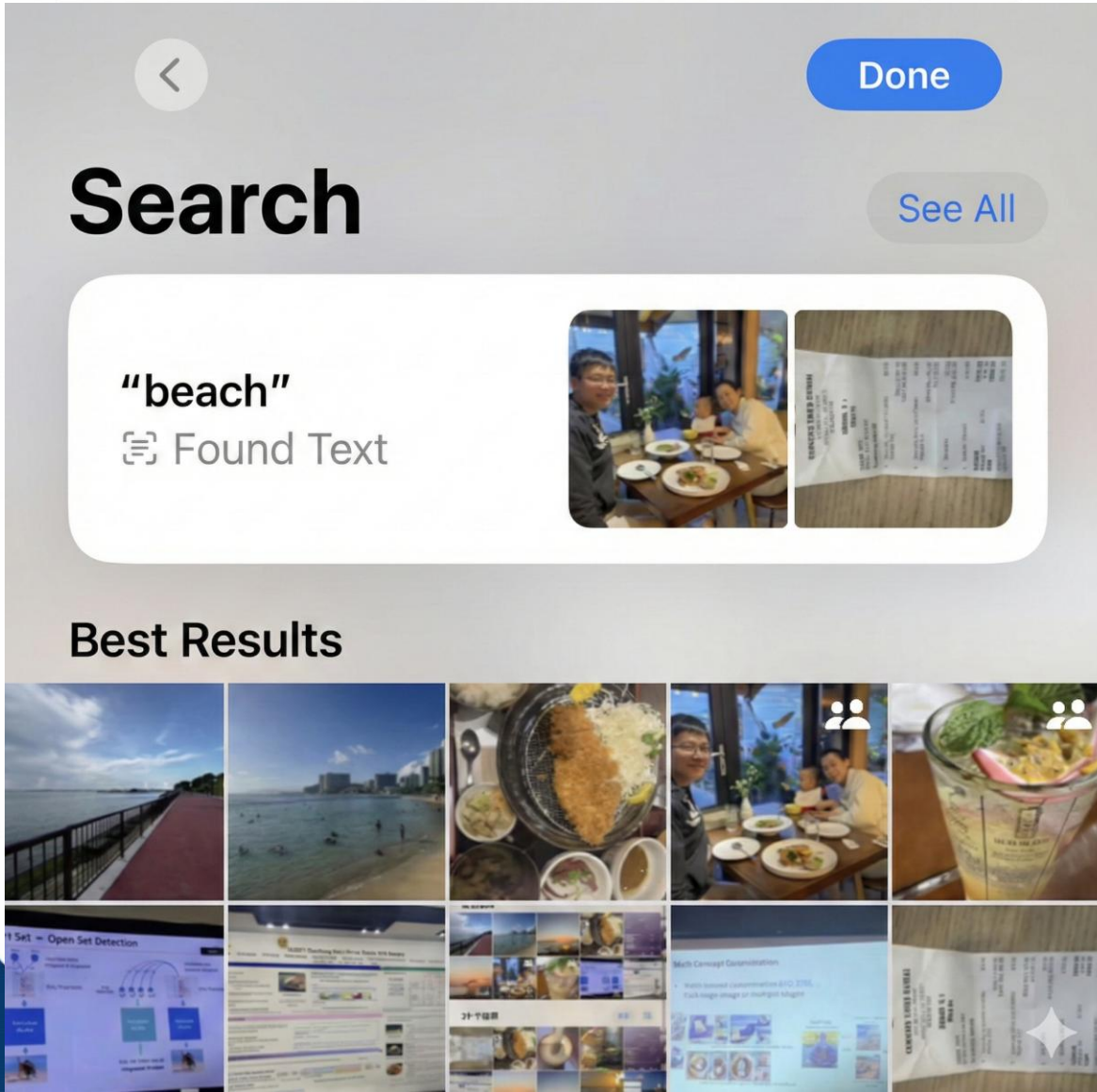
Let's Start



Open photos app in your phone and search something, e.g., Beach.



Smart Photo Album Search



Search: Beach → Results

No Manual Tags

AI Understands Photos

How AI Sees?

You: 🏖️ Beach



How AI Sees?

You: 🏖️ Beach



Computer: [123, 45, 67...]

[123]	45	67]	[16]	[44]	[45]	67]
67]	[100	20	50	[34]	[49]	50]
[100	50]	[100	20	[37]	[45]	50]
[100	[120	50]	[30]	[32]	45	67]
[200	[150]	25]	[30]	[45]	[30]	67]
[200	[150]	25]	[30]	[45]	[36]	[37]

How AI Sees?

You: 🏖️ Beach



Computer: [123, 45, 67...]

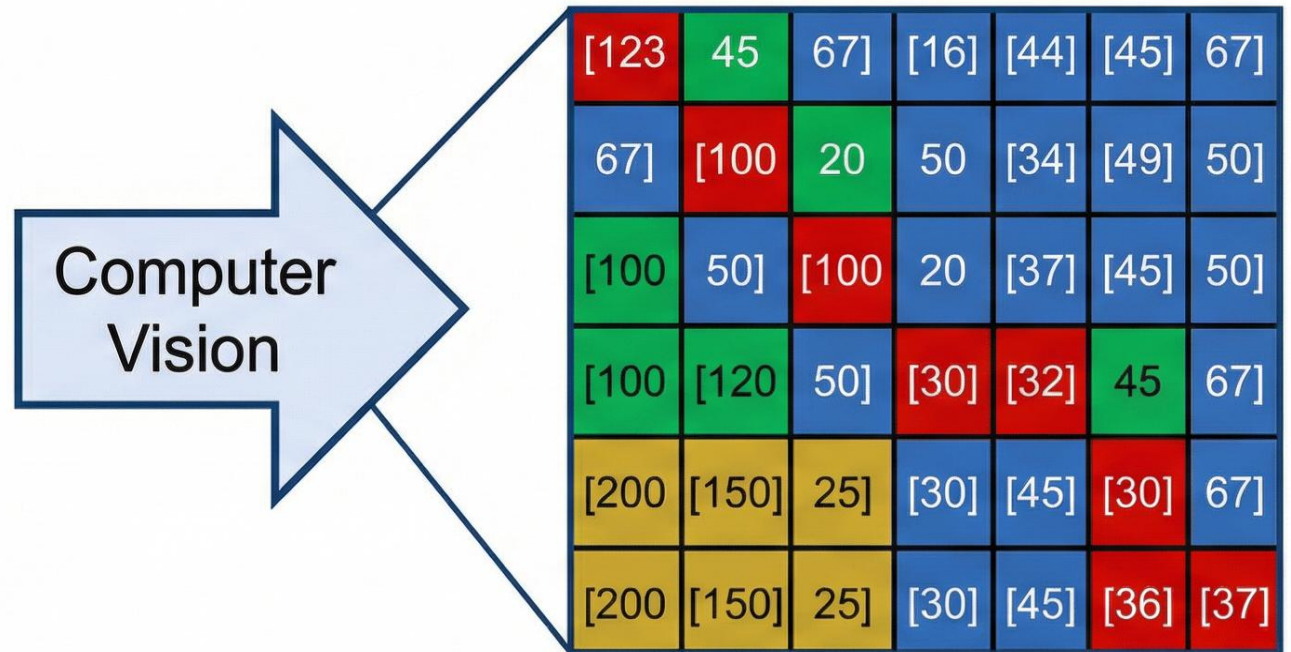


Image → Numbers → AI → Understanding

CV: Numbers → Understanding



Autonomous Driving



- Real-time:
 - 🚗 Vehicles
 - 🚶 Pedestrians
 - 🛣️ Lanes
 - 🚦 Signs

30+ FPS

Source: HyperChange

Beyond perception,
computer vision now can do **visual reasoning**!





Integration of Localization Evidence

Key Identifier Analysis

1. "You Yuan Cooked Food" Signage (Confidence: 0.44):

- Chinese and English names : 友源熟食 / YOU YUAN COOKED FOOD
- Typical naming style of Singaporean hawker center stalls

2. "Fine Met Cooked Food" Signage (Confidence: 0.41):

- Chinese and English names: 佳美熟食 / Fine Met Cooked Food
- Consistent with the characteristics of Singapore's multicultural food culture

3. "Fresh Squeezed Juice" Stall (Confidence: 0.38):

- Features a price tag "100" (possibly representing \$1.00)
- A common beverage stall found in Singaporean hawker centers

Key Evidence for Geographic Localization

1. Architectural Feature Verification:

- Metal ceiling structure
- Centralized dining area layout
- Ceiling fan ventilation system → Typical design of Singaporean Hawker Centers

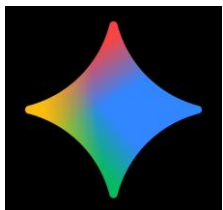
2. Cultural Identifiers:

- Bilingual Chinese and English signage
- Chinese cuisine such as roasted meats → Characteristics of Singaporean Chinese communities

3. Commercial Environment :

- Open-air dining space
- Centrally managed food stalls → Consistent with the Housing & Development Board (HDB) hawker center model





Final Positioning Conclusion

✅ **Confirmed Location:** Challenger Way Hawker Centre (Challenger Way 熟食中心) in Jurong (裕廊区), Singapore

✅ **Specific Characteristics:**

- Located on the ground floor of an **HDB** (组屋区) block
- An officially named **Hawker Centre** (小贩中心)
- Near **Jurong East MRT** station (邻近裕廊东地铁站)

📌 **Verification Suggestions:**

1. Search for "**Challenger Way Hawker Centre**" on Google Maps.
2. Check Street View to match the **metal ceiling structure** and **stall layout**.
3. Confirm whether stalls such as "**You Yuan Cooked Food**" (友源熟食) and "**Jia Mei Cooked Food**" (佳美熟食) exist.

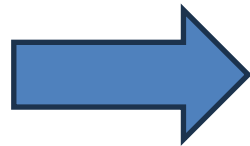


Image/3D/Video Generation





Input: A
squirrel eats
some super
spicy noddle



Scarlet Sentinel — Bloom of Destruction (猩红守望者 — 毁灭之华)

Structure (Variable edit duration; total length 25s; content 23.00s + black screen 2.00s; Mode: Standard; Rhythm factor: 2.0)

[#1 0.05s]

- ACTION:** Static shot — a knight kneels on one knee, supporting herself with a glowing sword, surrounded by red flowers.
- CAMERA:** Medium shot, fixed, centered.
- SUBJECT:** Female knight; iron mask beneath the hood; dull golden armor; dust-covered cloak; translucent red light blade; kneeling posture.
- SCENE:** Ruined underground temple cave; floor covered in red flowers; dust floating in the air.
- LIGHT:** Cold light shines down from crevices in the vault; the sword acts as a red light source; ethereal mist.
- CAM:** Medium shot; centered symmetry; 35mm texture; locked camera position.

[#2 1.00s]

- ACTION :** She plunges the sword deeper into the ground; humming intensifies; petals tremble.
- CAMERA:** Push in from a low angle at petal height. — — **[Dynamic] /***
Qi/Aura */
- SUBJECT:** Armored hand tightly gripping the sword hilt; iron mask lowered.
- SCENE:** Center of the sea of flowers.
- LIGHT:** Red core halo; soft rim lighting.

...



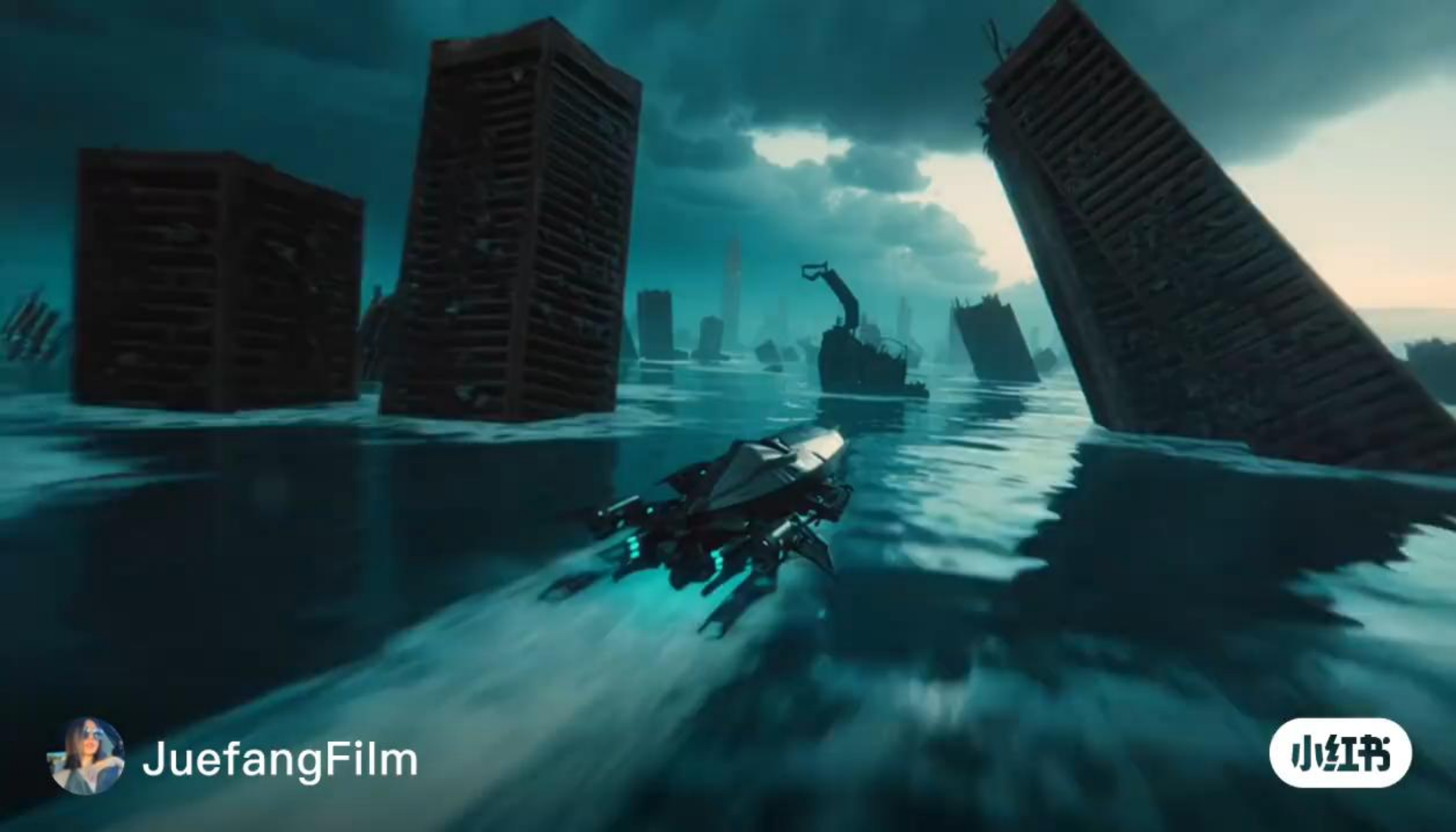
This field is full of opportunities.

And most opportunities belong to young people,
like you









JuefangFilm

小红书

There are still a lot of projects that could be commercialization...

After taking this class, contact me if are interested.



Part 1

What is Computer Vision?



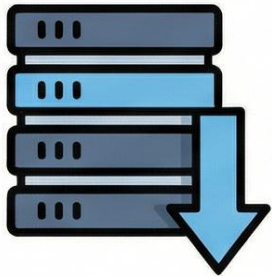
Computer Vision: Definition



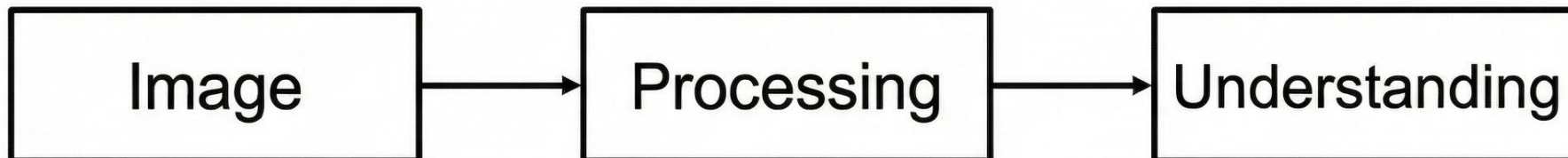
Enable computers to **SEE** and **UNDERSTAND** visual world



EXTRACT meaningful information

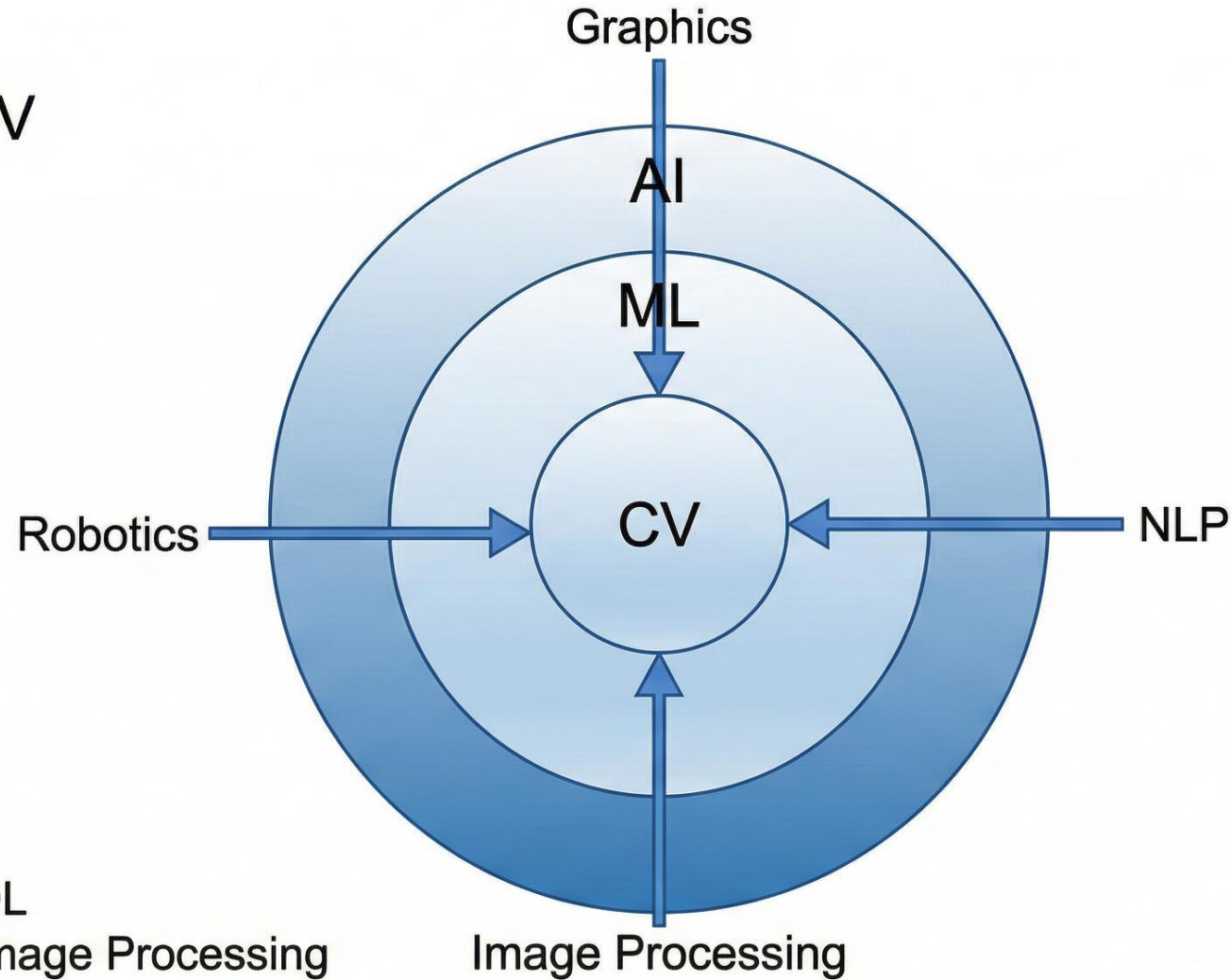


Goal: Human-level **PERCEPTION**



CV in Context

$AI \supset ML \supset CV$



CV uses: ML, DL

CV builds on: Image Processing

CV relates to: NLP, Robotics



Main CV Tasks



Classification:
What?

Main CV Tasks



Classification:
What?



Detection:
Where?

Main CV Tasks



Classification:
What?



Detection:
Where?



Generation:
Create?

Main CV Tasks



Classification:
What?



Detection:
Where?



Generation:
Create?



Segmentation:
Boundaries?

Main CV Tasks



Classification:
What?



Detection:
Where?



Generation:
Create?



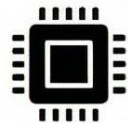
Segmentation:
Boundaries?

+ More: Pose, Depth, Tracking...

<https://huggingface.co/spaces/Ruicheng/MoGe-2>



Quick Recap

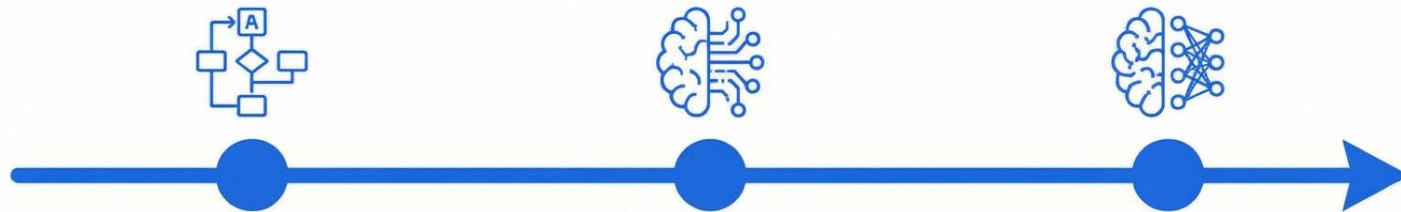
- ✓ CV = Subdomain of AI  
- ✓ Tasks: Understanding  → Generation 
- ✓ Applications: Everywhere   

Next: History →

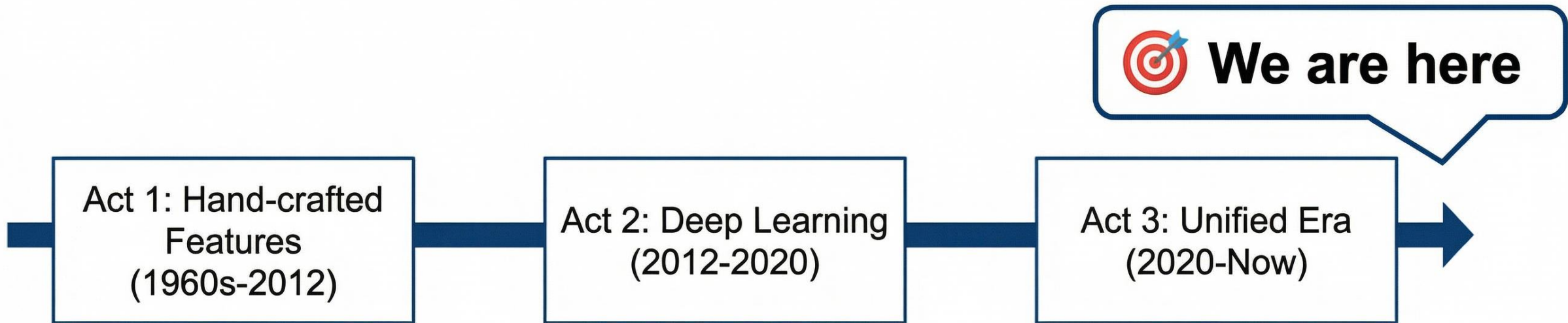


Part 2

History of Computer Vision A Journey Through Three Eras



Three Eras of CV



Act 1

Hand-crafted Features Era

1960s – 2012



MIT Summer Vision Project (1966)

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

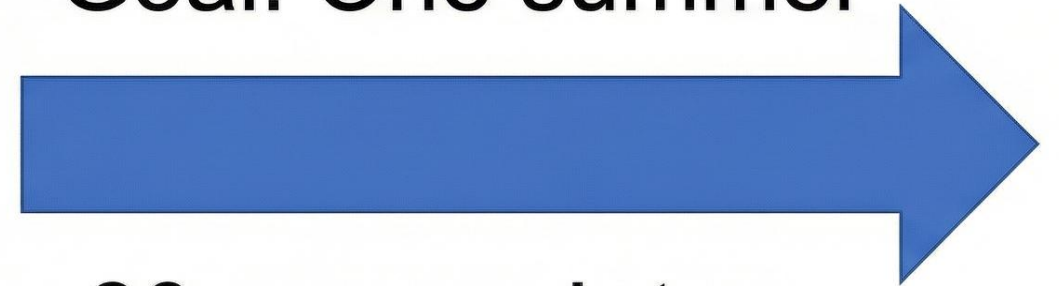
July 7, 1966

THE SUMMER VISION PROJECT

Seymour Papert.

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition!!.

Goal: One summer

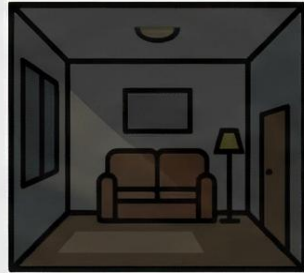


60+ years later...

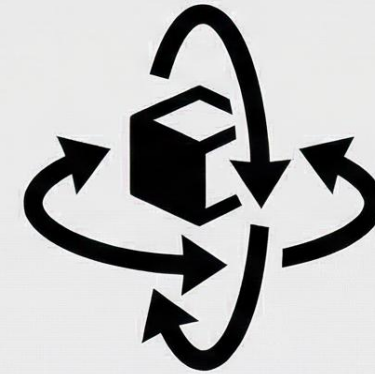
Lesson: Vision is HARD



Why is Vision So Hard?



Lighting



Viewpoint



Occlusion

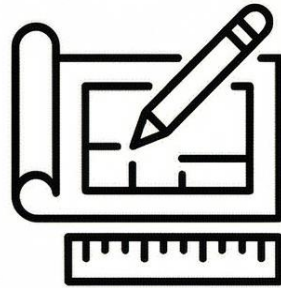
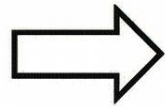


Clutter

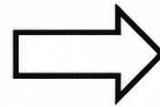
Hand-crafted Approach



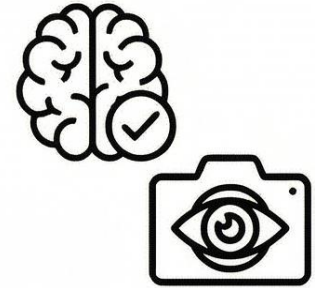
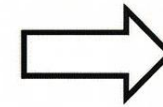
Human Expert



Design Rules

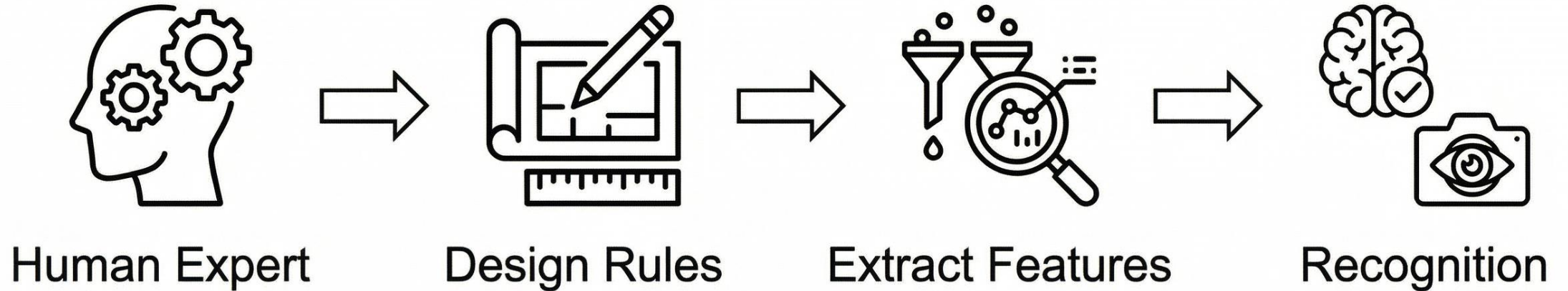


Extract Features



Recognition

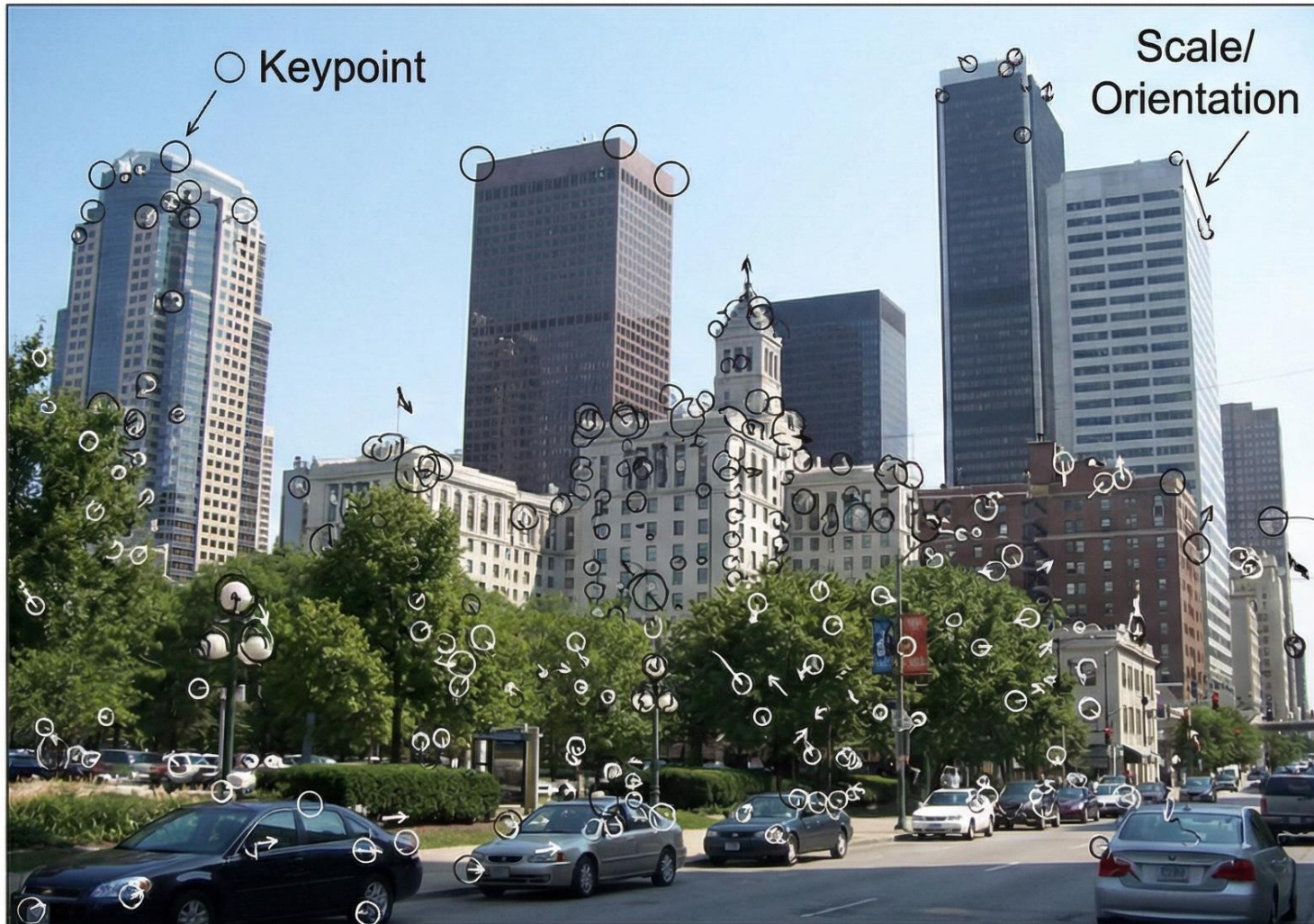
Hand-crafted Approach



Philosophy: Humans tell machines

Examples: SIFT, HOG, Haar

SIFT: Feature Detection



	Scale-invariant
	Rotation-invariant
	Human-designed

SIFT Features

Detects: Keypoints (corners, edges)

Invariant to: Scale, Rotation
Human-designed

Limitations

✗ Task-specific

✗ Poor generalization

✗ Hard to design

✗ Performance ceiling

Breakthrough needed...



Act 1 Summary

1960s-2012
50+ years



Human-designed
rules

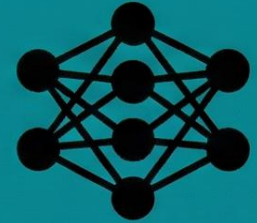


Basic CV
applications



Task-specific
limitations

2012
AlexNet



Then everything changed...

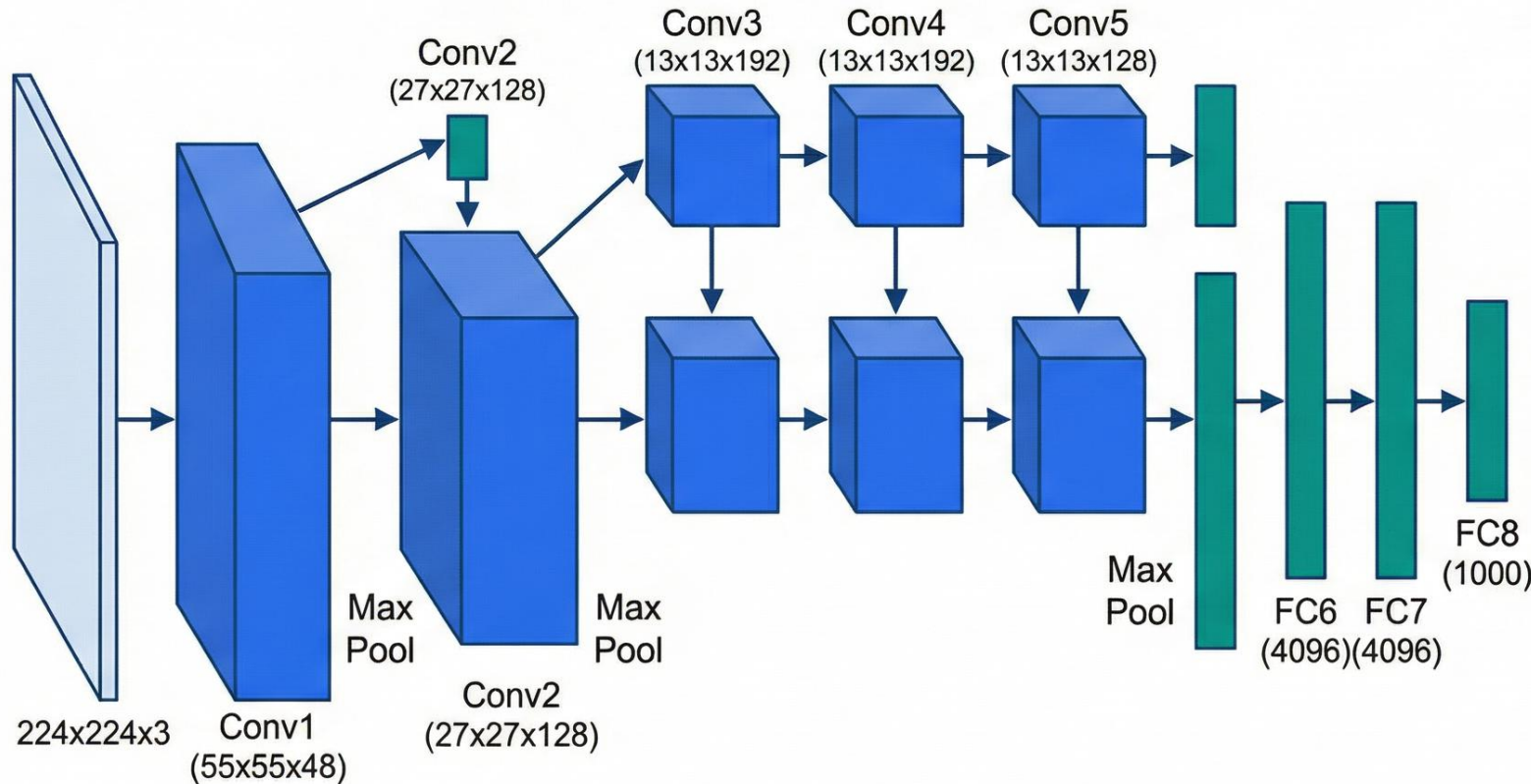
Act 2

Deep Learning Era

2012 – 2020



AlexNet (2012)



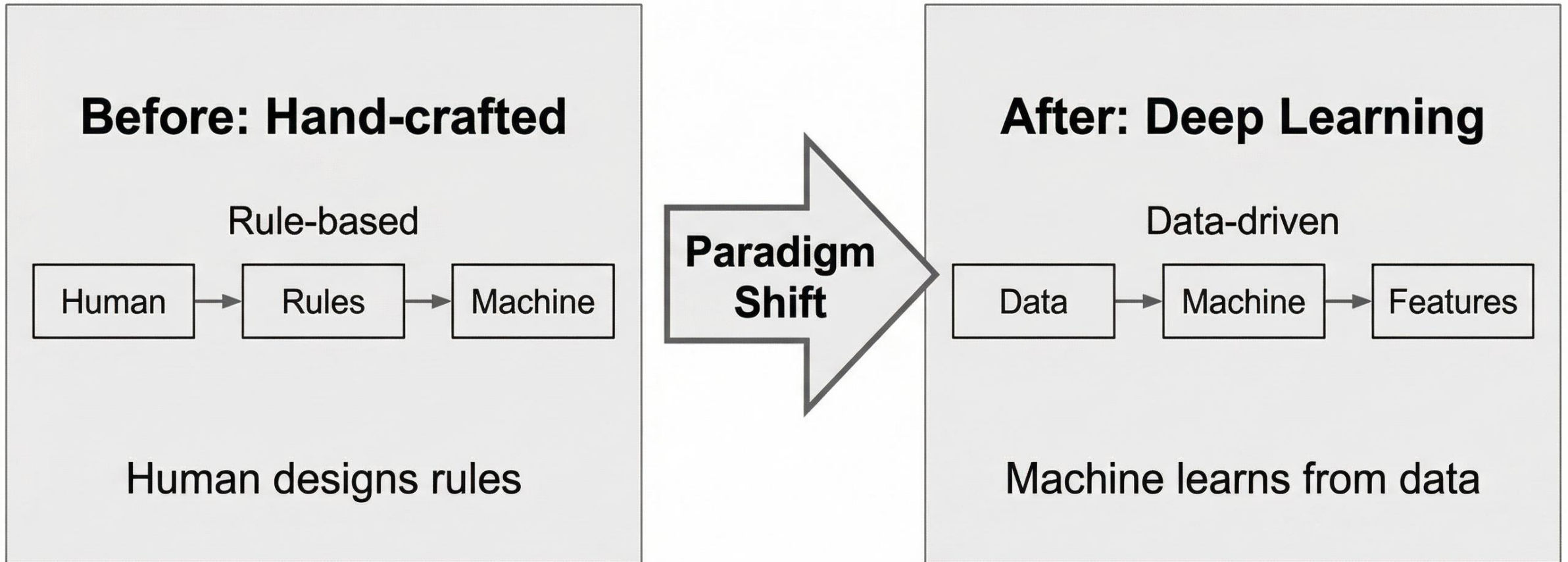
ImageNet Challenge:
AlexNet: 15.3%
2nd Place: 26.2%

Gap: 10%
Everyone Stunned

The Game Changed

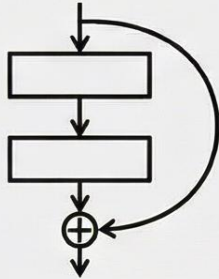


Paradigm Shift



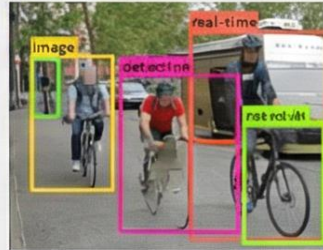
Tell what → Learn how

DL Achievements (2012-2020)



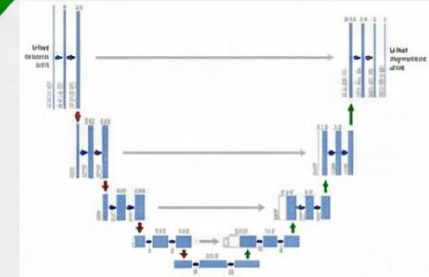
ResNet:

Classification: 95%+



YOLO:

Detection: Real-time



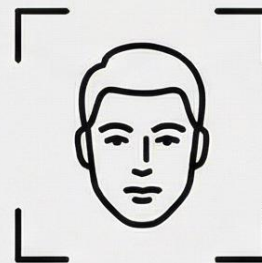
U-Net:

Segmentation: Pixel-level



OpenPose:

Pose: Joints



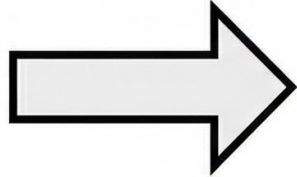
FaceNet:

Face: Beyond human

+ Depth, Video, 3D...

Act 2 Limitations

- ✗ One model, one task
- ✗ Understanding $\neq$ Generation
- ✗ Massive labeled data
- ✗ Lacks generalization

Can we unify them? 

Act 3

Unified Era



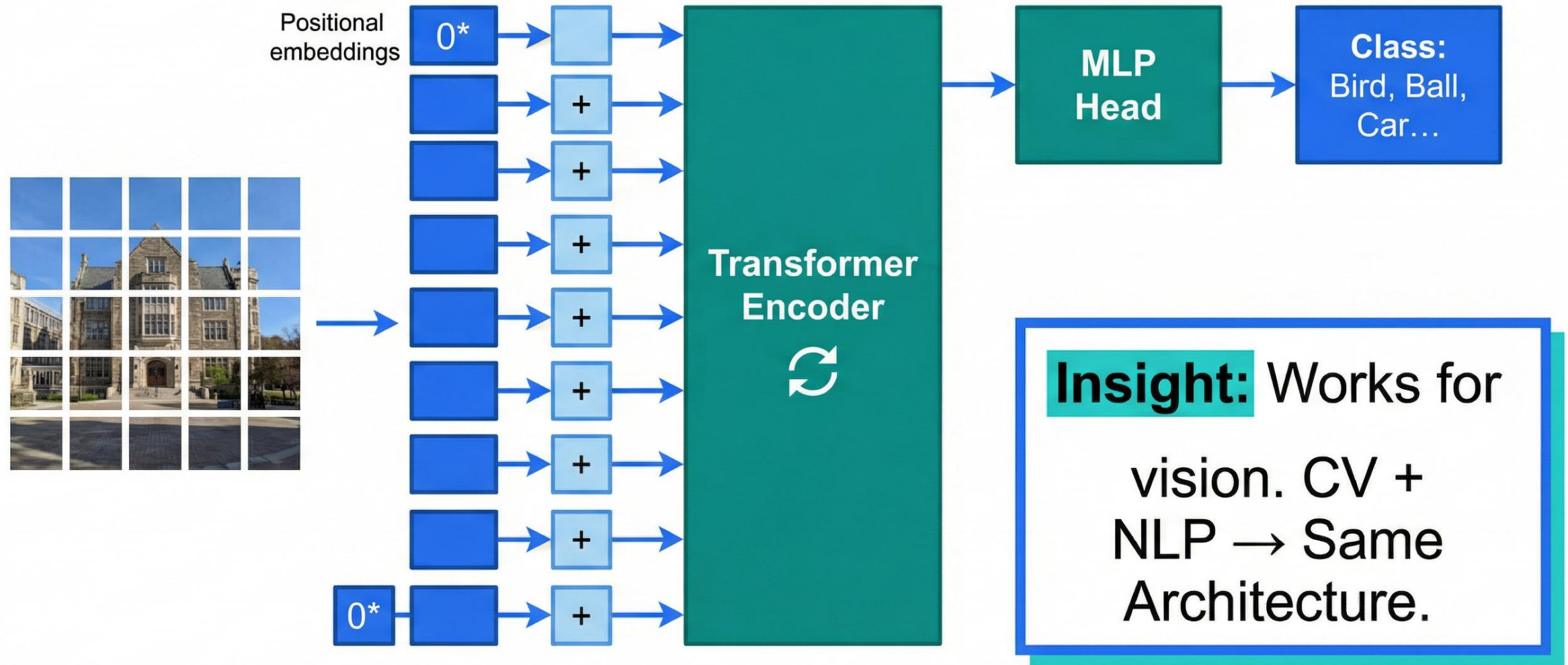
2020 – Now



We are here



Vision Transformer (2020)



Four Unifying Trends

1 Multimodal
Text + Vision

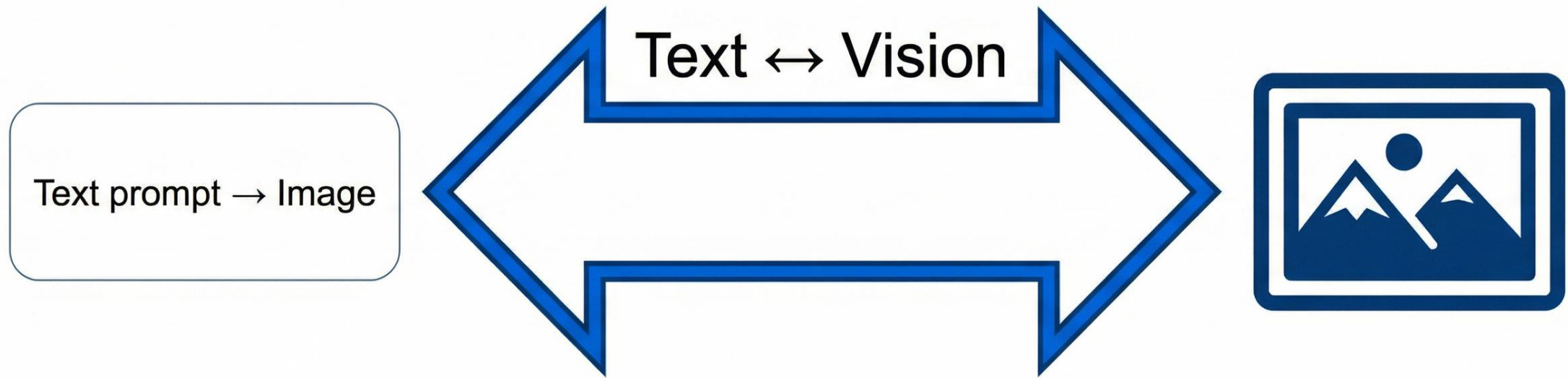
2 Zero-shot
No training

3 Visual Dialog
Reasoning

4 World Model
Unified

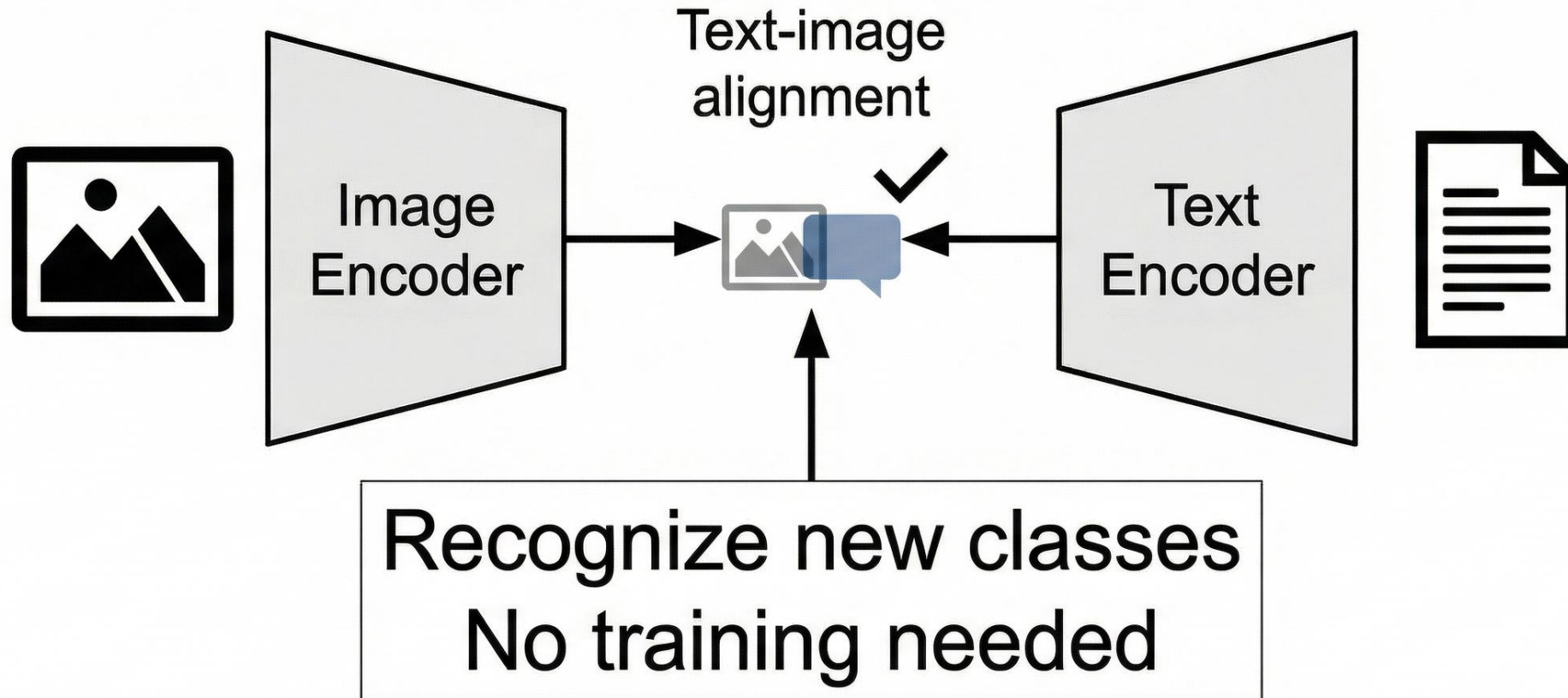


Trend 1: Multimodal



DALL-E, Midjourney, Stable Diffusion

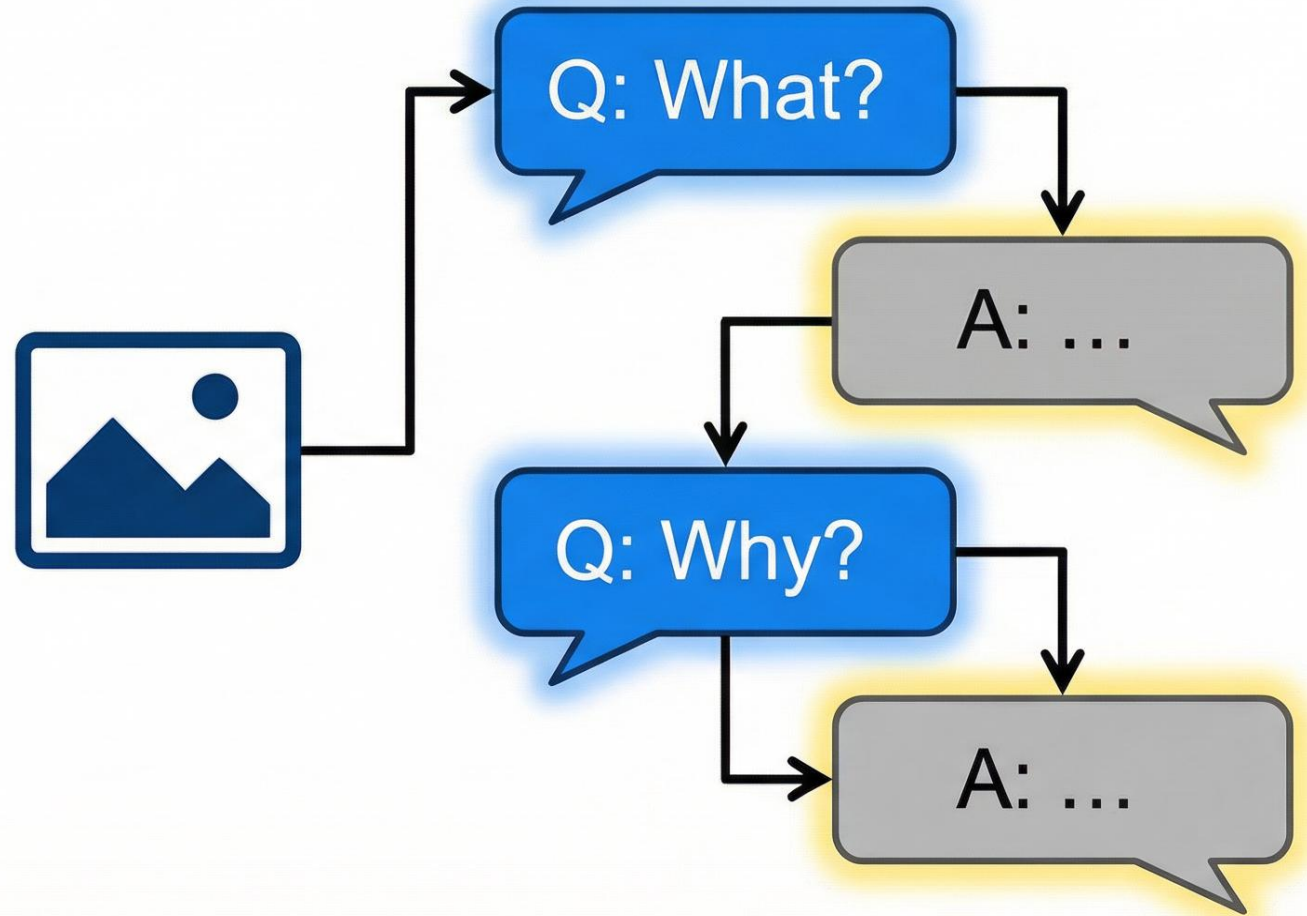
Trend 2: Zero-shot



Example: CLIP (OpenAI)

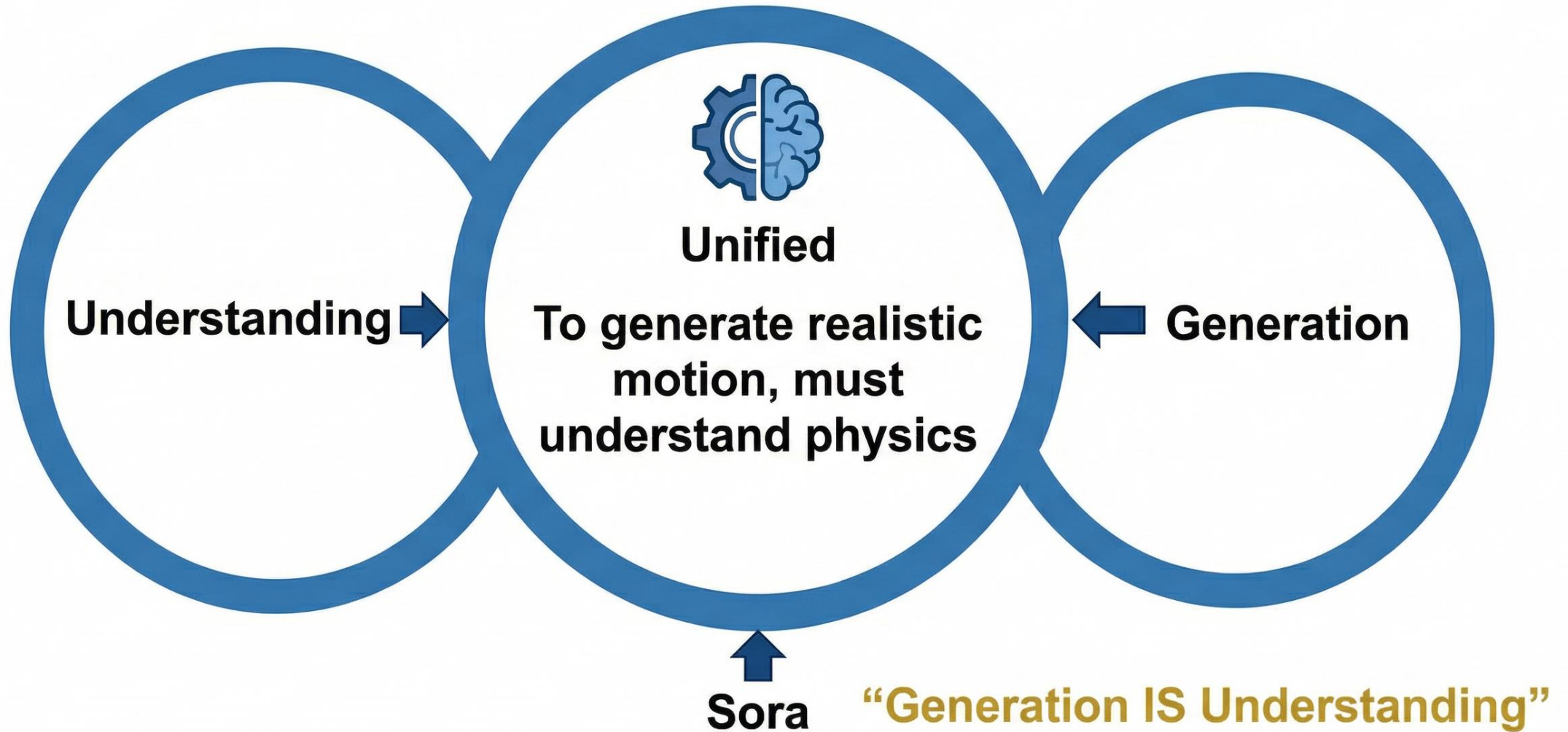
Trend 3: Visual Dialog

- Answer any question
- Multi-turn reasoning
- From recognition → understanding

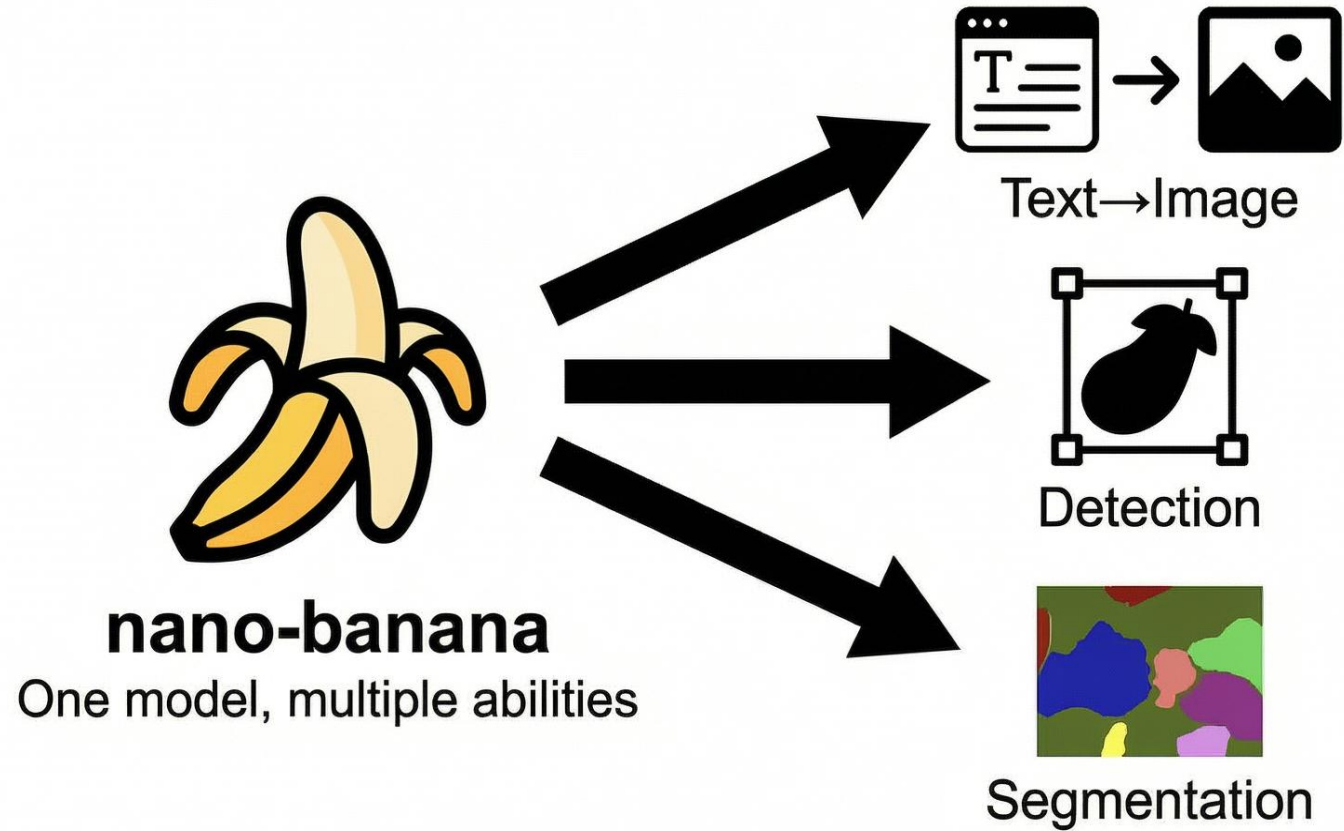


GPT-4V, Gemini, Claude

Trend 4: World Model



Live Demo: nano-banana



Let's see it!





detect all faces with red boxes

◆ 显示思路 (Nano Banana Pro) ▼



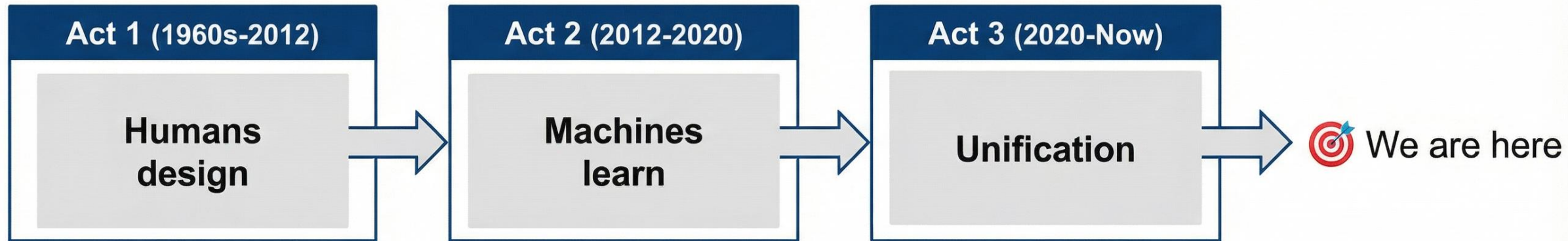
Please use different colors to segment all person in the image.

显示思路 (Nano Banana Pro) ▼





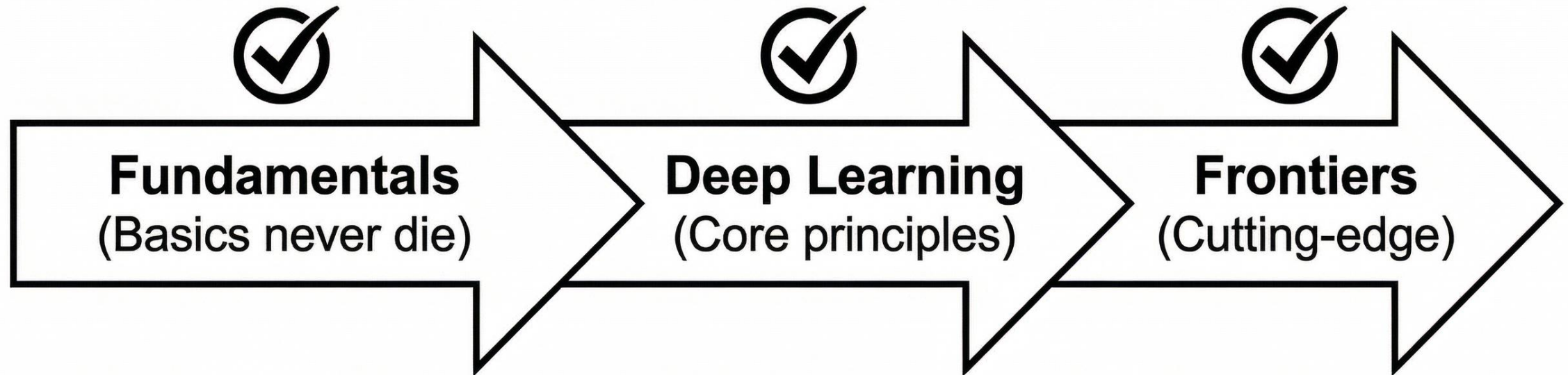
Three Eras: Complete Timeline



Each builds on last



Why Fundamentals?



- ✓ Basics never die
- ✓ Understand principles
- ✓ DL builds on them

Weeks 1-4 → 5-9 → 10-12

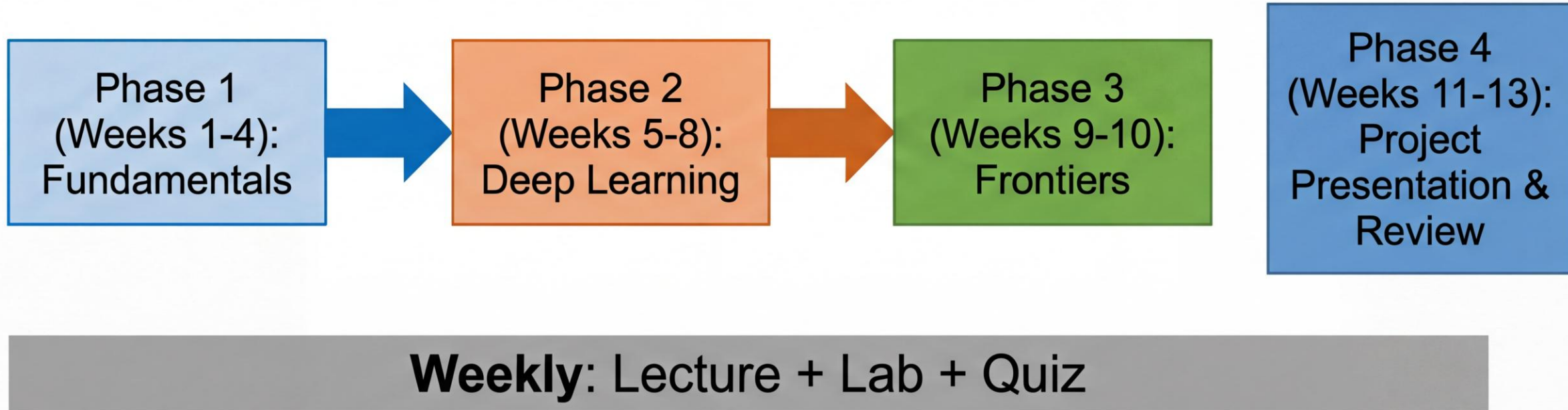
Course: Fundamentals → DL → Frontiers

Part 3

Course Overview



Course: 13 Weeks



Phase 1: Fundamentals

Week 1: Intro & Image Basics

Week 2: Image Filtering

Week 3: Edge Detection

Week 4: Feature Extraction

Goal: Build foundation



Phase 2: Deep Learning

Week 5: Machine Learning Basics for Vision

Week 6: Neural Networks (MLP) & CNN Basics

Week 7: Training Strategies & Data

Goal: Master deep learning based CV methods.



Phase 3: Frontiers

Week 8: Object Detection & Segmentation

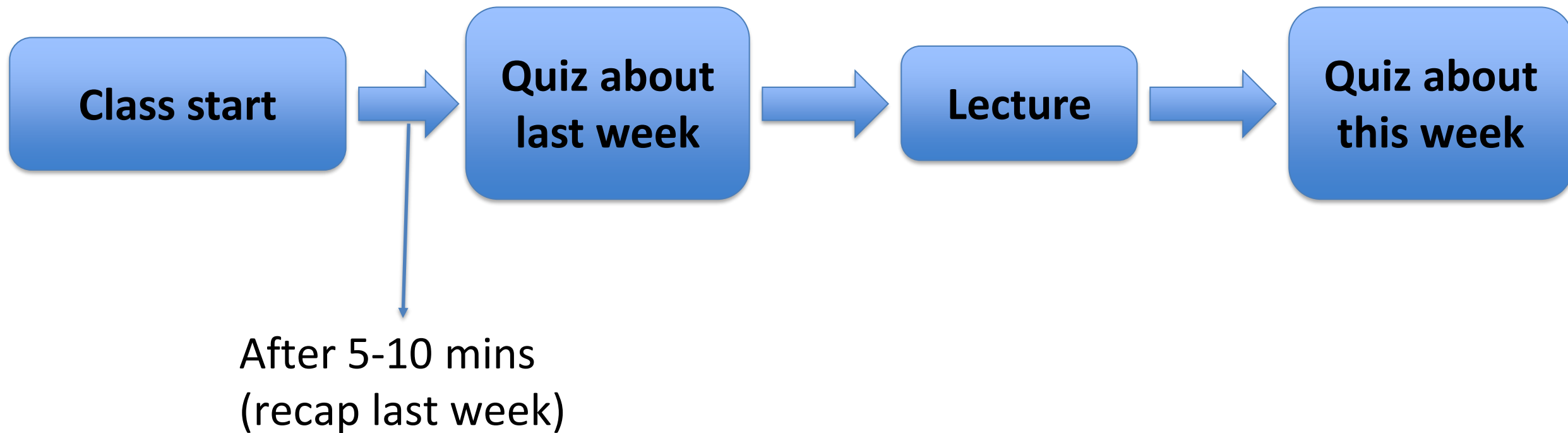
Week 9: Vision Transformers (ViT)

Week 10: Generative AI & Frontiers



Goal: Explore cutting-edge CV methods.

Weekly Structures



The first 5–10 minutes are reserved to allow late arrivals to check in..

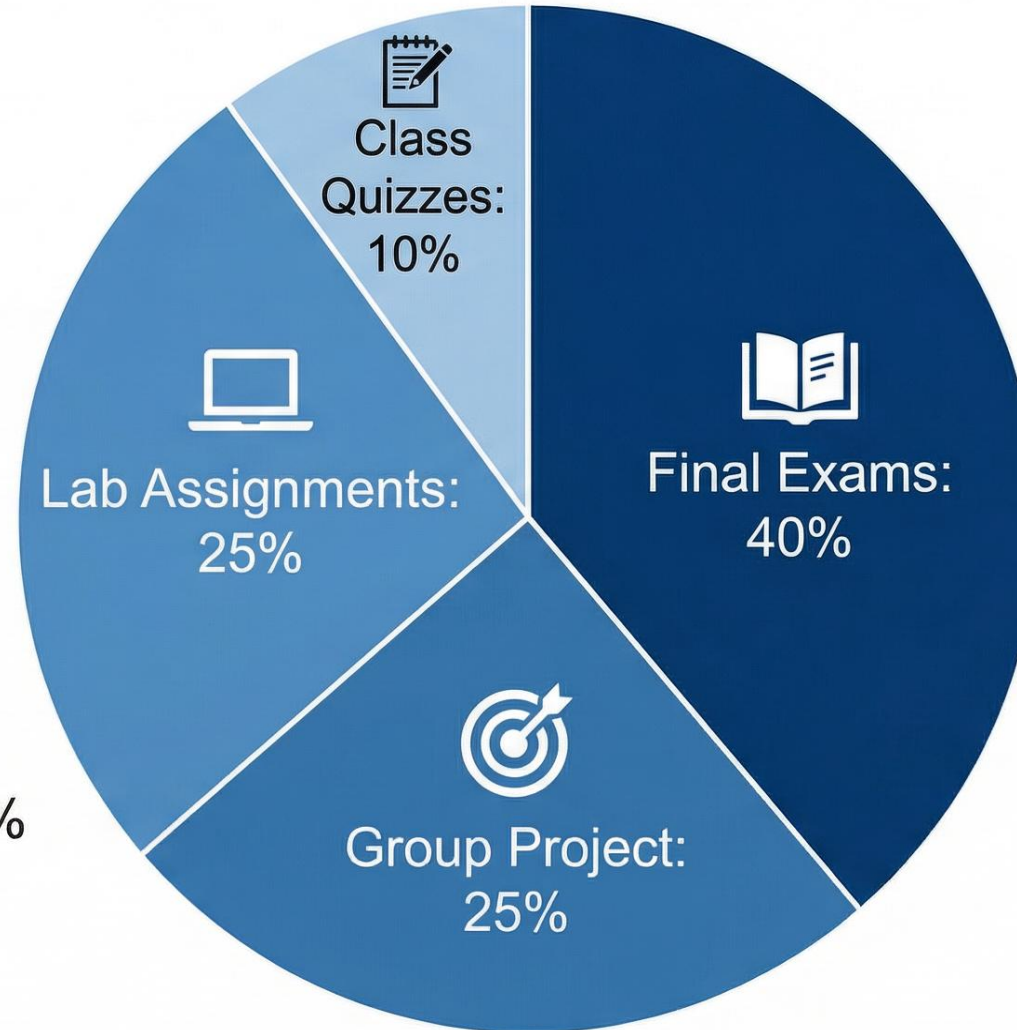


Part 4

Grading & Assessment



Grading Breakdown



-  Class Quizzes: 10%
-  Lab Assignments: 25%
-  Group Project: 25%
-  Final Exams: 40%

Total: 100%



Class Quizzes: 10%

Frequency:	Almost weekly
Format:	MC, 5-10min
Scoring:	Average all (0 for absence)



Purpose: Maintain
pace, identify gaps
(not penalize)

Lab Assignments: 25%

Weekly (~12 labs)
Pass/Fail (Complete = Pass)
Scoring: # Pass / Total

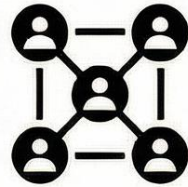
No pressure! Just complete.



Group Project: 25%



Team: 2 students



Week 3-4:
Team
Formation



Week 5-6:
Topic
Selection



W12 & W13:
Presentation



Content: CV
task
implementation

Grading Dimensions

- Implementation
- Innovation
- Presentation



Assessment Breakdown



Class Quizzes: 10%



Lab Assignments: 25%



Group Project: 25%



Final Exams: **40%**

Total: 100%

Balanced: 60% continuous + 40% final



Academic Integrity



Lab

Discuss

OK, own code



Team

OK, no copying



Quiz/Exam

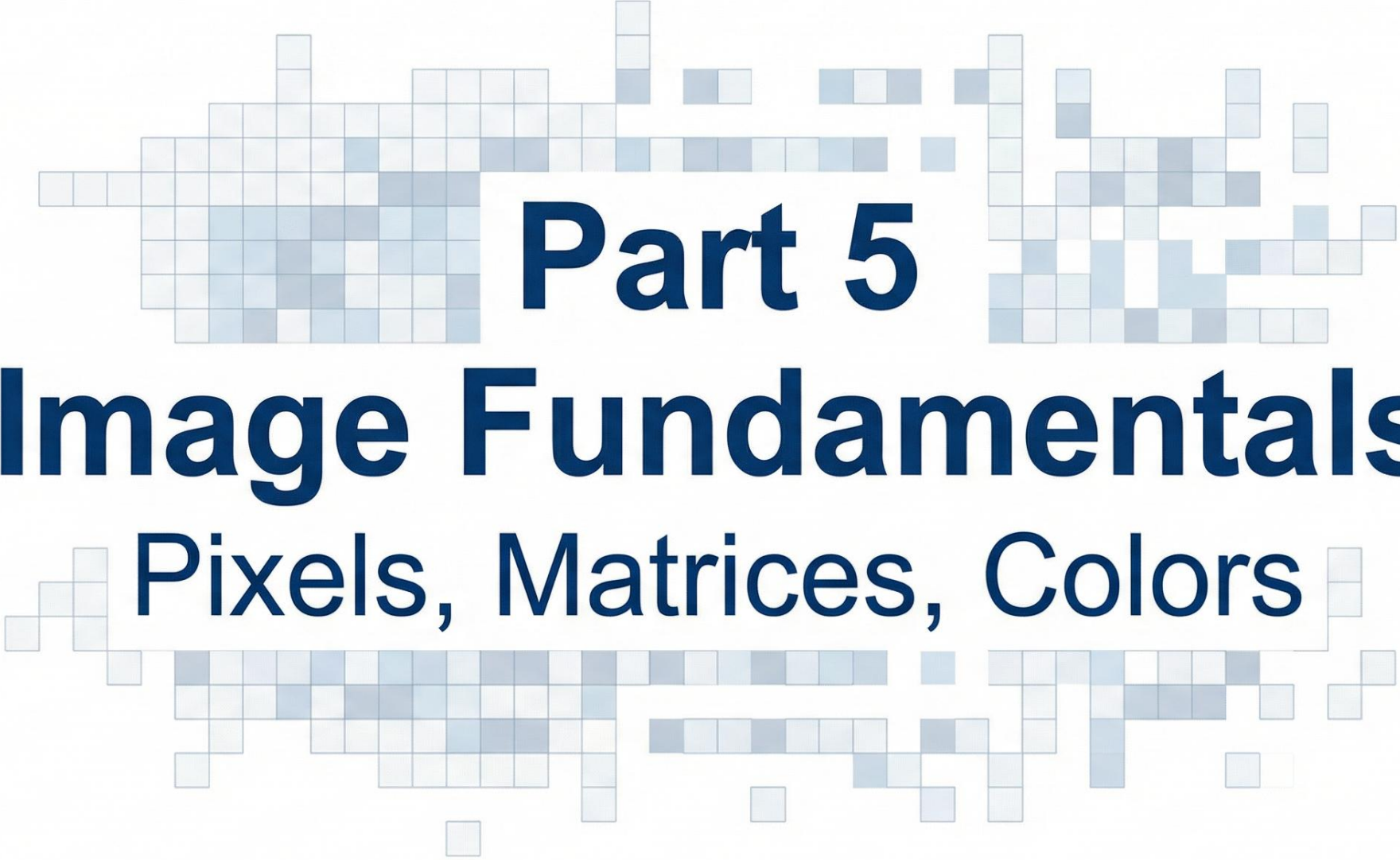
Individual work



Violation: Serious consequences

Be honest, be fair.





Part 5

Image Fundamentals

Pixels, Matrices, Colors

What Computer Sees?



Gap

[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[16, 61, 32]	[46, 61, 69]	[46, 61, 39]	[123, 45, 67]	[123, 45, 100]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[16, 49, 35]	[46, 61, 66]	[49, 49, 40]	[123, 45, 67]	[123, 45, 100]
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[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[169, 167, 22]	[188, 163, ...]	[80, 160, 22]	[108, 168, ...]	[123, 45, 100]
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[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22]	[40, 93, 21]	[90, 92, 21]	[32, 63, 24]	[37, 72, 18]	[123, 45, 106]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22]	[90, 91, 21]	[89, 168, 22]	[59, 93, 64]	[123, 46, 42]	[123, 46, 108]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[89, 160, 21]	[59, 160, 21]	[59, 99, 24]	[123, 46, 57]	[122, 46, 108]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22]	[89, 169, 22]	[59, 163, 22]	[39, 59, 21]	[122, 46, 55]	[122, 46, 991]
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[123, 45, 67],	[200, 10, 59],	[89, 167, 22],	[89, 167, 22]	[99, 167, 29]	[80, 99, 26]	[76, 52, 31]	[123, 46, 61]	[123, 46, 201]
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[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22]	[89, 99, 29]	[80, 183, 31]	[80, 97, 33]	[120, 45, 67]	[123, 45, 221]

You: Beach, Waves, Sunshine, Happiness

Computer: [123, 45, 67]...

This gap = where CV begins



Image = Matrix



=

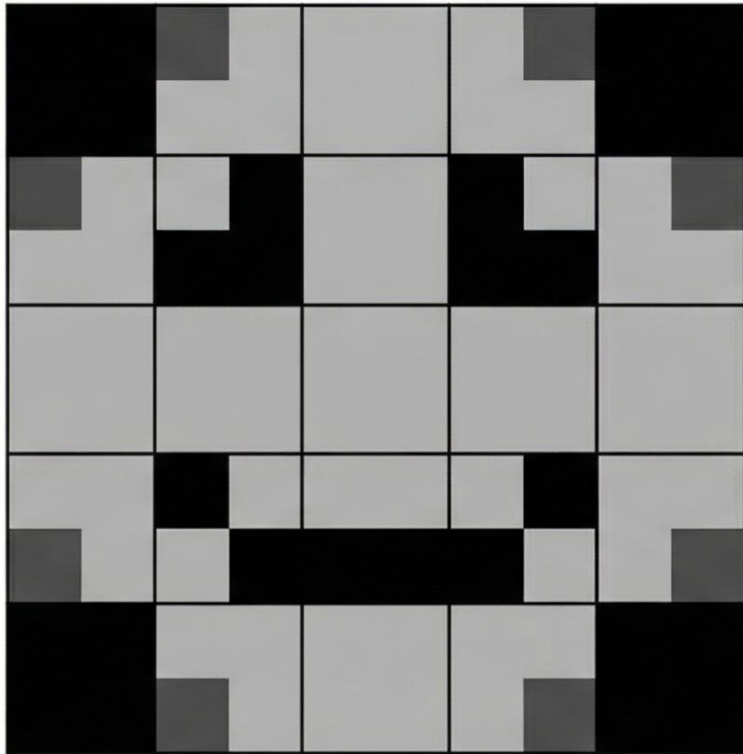
255	200	255
200	50	200
255	200	255

Each pixel = number
(grayscale) or 3
numbers (color)

Range: 0-255

Everything is
numbers.

Grayscale Example



Single channel:
1 number per
pixel.

0 = Black.
255 = White.

5x5 matrix.

0	255	255	0	0
0	255	113	255	0
0	255	113	255	0
0	255	113	255	0
0	0	0	0	0

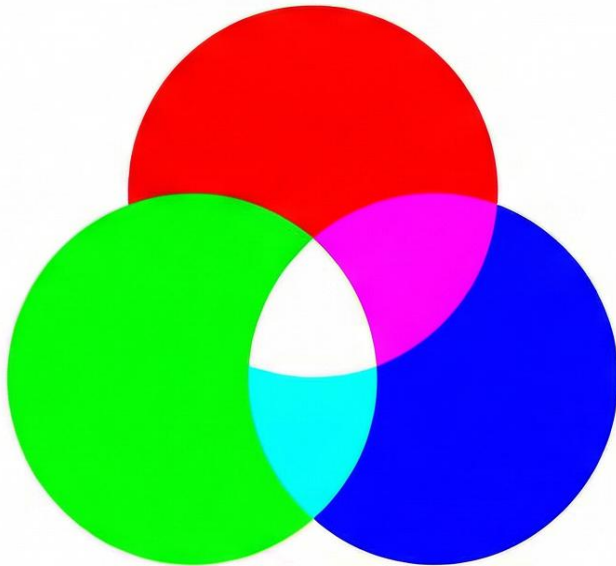
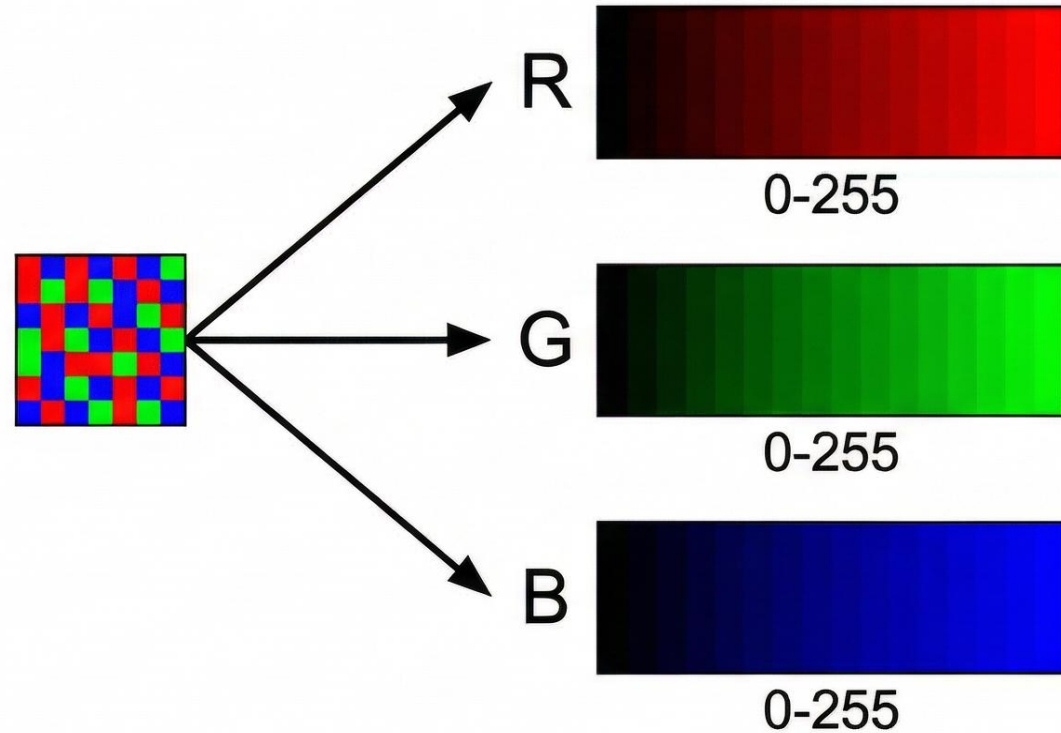
0 (Black)



→ 255 (White)

Color = 3 Channels (RGB)

Color = 3 Channels
Each pixel = (R, G, B)
Each: 0-255

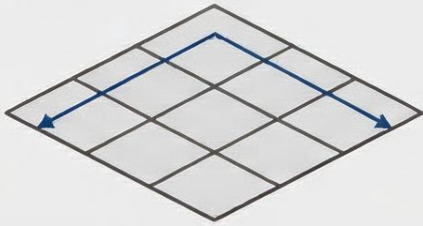


→ Red + Green + Blue → All colors

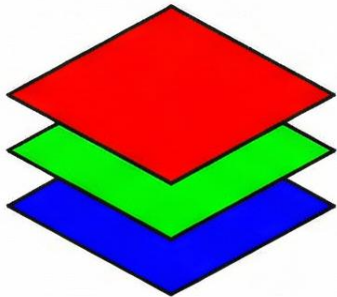
Dimensions & Resolution

Dimensions

Grayscale: $H \times W \times 1$



Color:
 $H \times W \times 3$

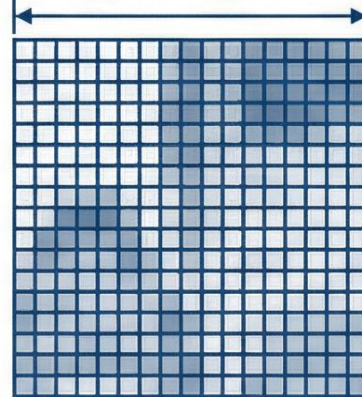


Resolution: Total pixels

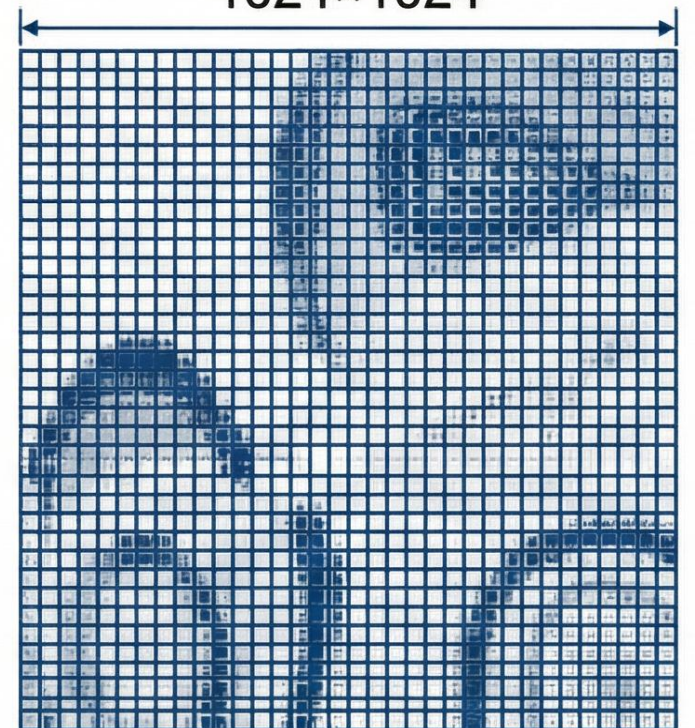
64×64



256×256



1024×1024



Higher resolution → More details, Larger file

Demo Code: Load & Inspect

```
import cv2

img = cv2.imread('photo.jpg')
print(img.shape)
# (1080, 1920, 3)

pixel = img[100, 200]
print(pixel)
# [B, G, R]
```

OpenCV uses BGR order



Demo Code: Split Channels

```
b, g, r = cv2.split(img)
print(b.shape)
# (1080, 1920)
```

```
cv2.imshow("R", r)
cv2.imshow("G", g)
cv2.imshow("B", b) '
```

R



G

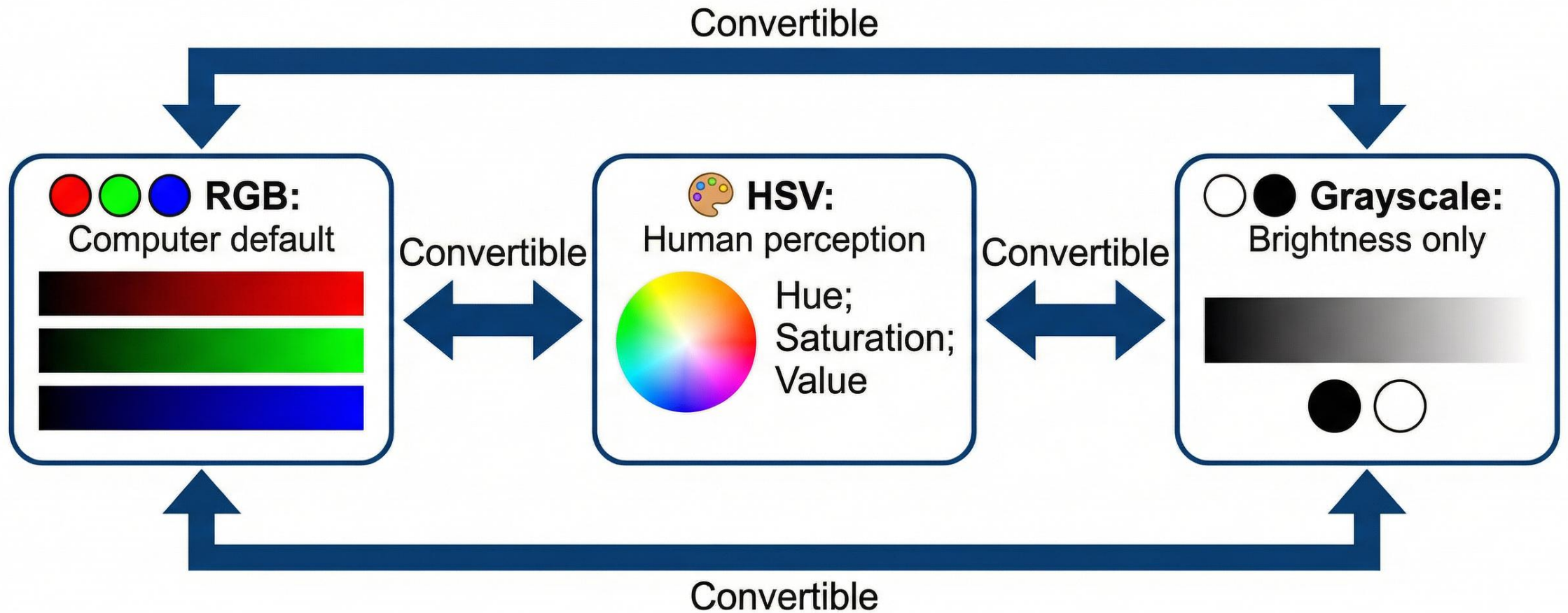


B



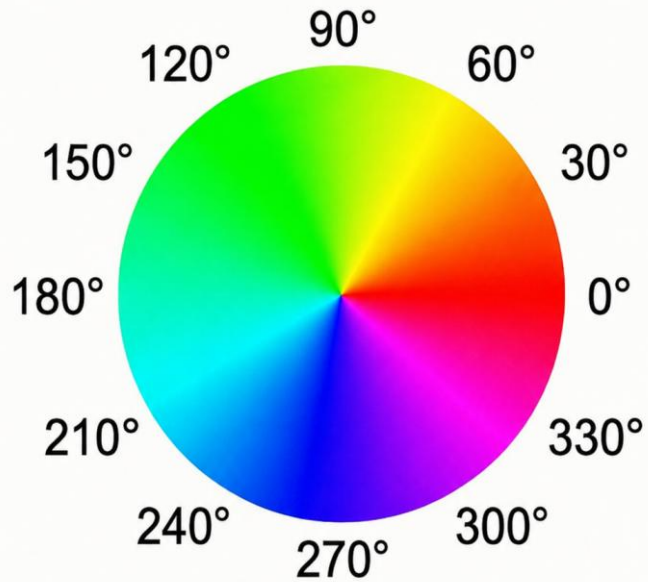
Each = grayscale.

Color Spaces



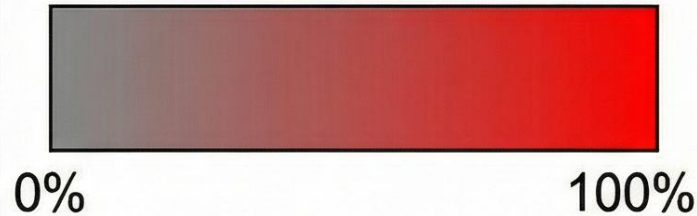
HSV Color Format

HUE (H): Color Type



0-360°: Defines base color
(e.g., Red, Blue, Green).

SATURATION (S): Color Purity



0-100%: Intensity.
Low=Gray, High=Vibrant.

0-100%: Intensity.
Low=Gray, High=Vibrant.

VALUE (V): Brightness



0-100%: Lightness.
Low=Black, High=White.

0-100%: Lightness.
Low=Black, High=White.



Color Space Conversion

```
# RGB -> HSV
# Logic: Project RGB Cube to Cylinder (V=Max, S=Range,
H=Angle)
V, S, H = max(R, G, B), (max(R, G, B) - min(R, G, B)) / max(R, G, B),
get_angle(R, G, B)

hsv = merge(H, S, V)

# RGB -> Gray
# Logic: Weighted Sum based on Eye Sensitivity
(Green>Red>Blue)

gray = 0.114 * Blue + 0.587 * Green + 0.299 * Red

print(gray.shape) # (H, W)
Grayscale has 2D only
```



Color Space Conversion

```
# RGB → HSV
hsv = cv2.cvtColor(img,
cv2.COLOR_BGR2HSV)

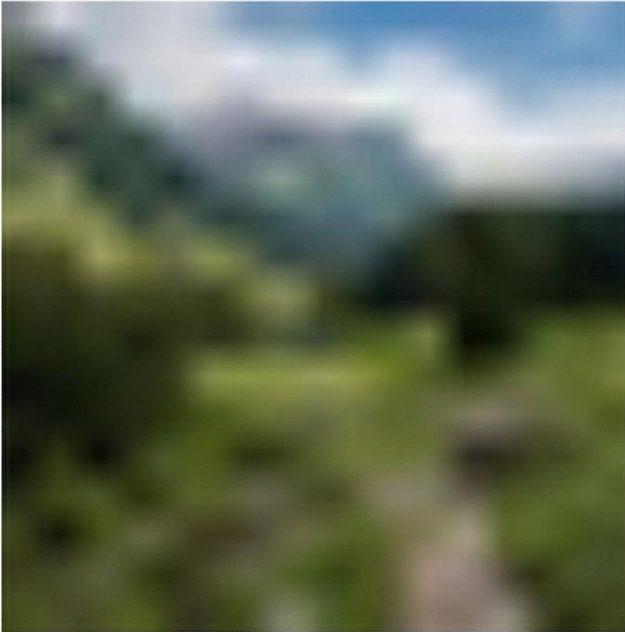
# RGB → Gray
gray = cv2.cvtColor(img,
cv2.COLOR_BGR2GRAY)

print(gray.shape)
# (H, W)
Grayscale has 2D only
```



Resolution Matters?

Resolution Impact



Low (32x32): Fewer details,
Smaller, Faster



High (1024x1024): More
details, Larger, Slower

Trade-off: Task needs vs Resources



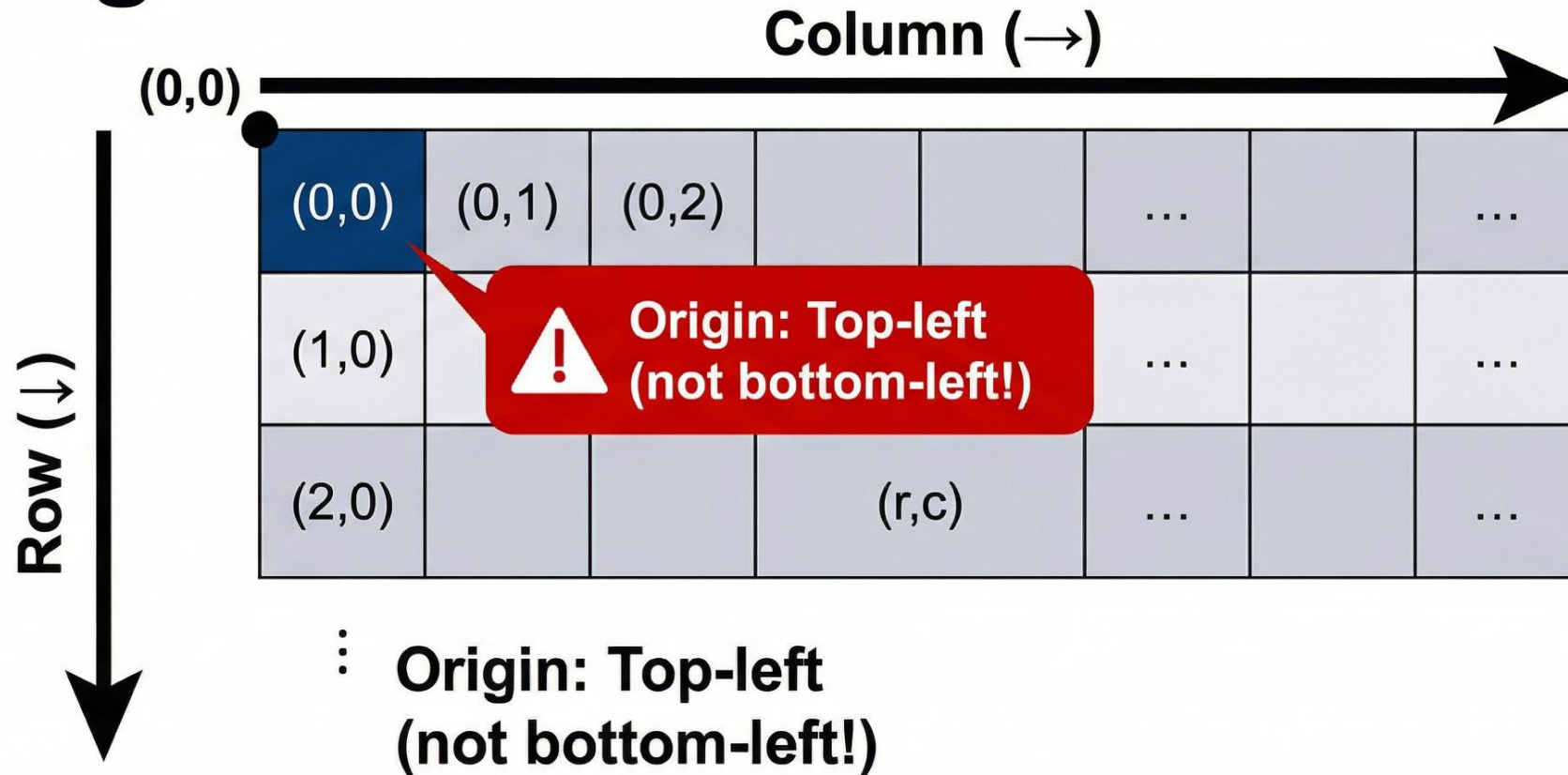
Image Fundamentals: Summary

- ✓ Image = Matrix of numbers
- ✓ Grayscale: $H \times W \times 1$ (0-255)
- ✓ Color: $H \times W \times 3$ (RGB)
- ✓ Color spaces convertible
- ✓ Resolution = pixel count

“Everything is numbers”



Image Coordinates



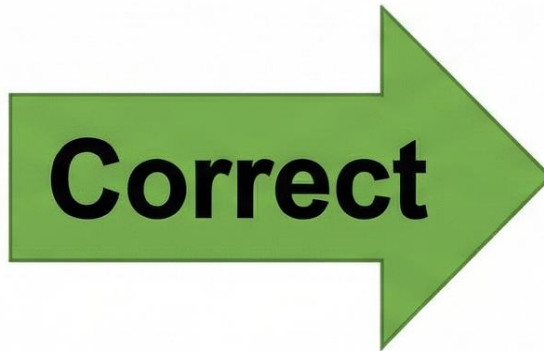
First index: Row (↓)
Second index: Col (→)

`img[row, col]`
`= img[y, x]`

Common Mistake

✗ Wrong:

`img[x, y]`



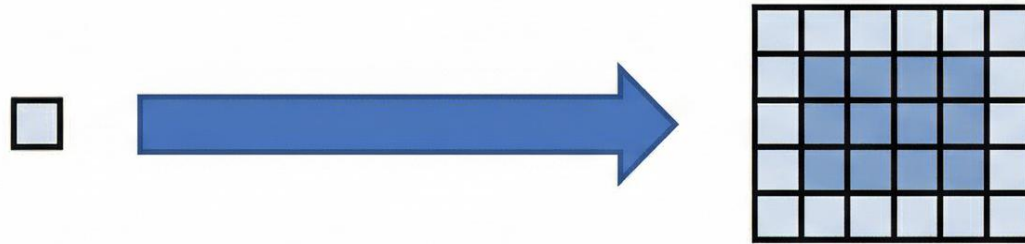
✓ Correct:

`img[y, x]` ✓
`img[row, col]`

Memory: `y` = vertical (row), `x` = horizontal (col)

Next Week: Image Filtering

Concept: Pixel-wise $\rightarrow$ Neighborhood



Topics: Blur, Sharpen, Denoise, Convolution



Lab: Implement filters



See you!

Useful information

- Top conferences
 - Computer Vision and Pattern Recognition (CVPR)
 - International Conference on Computer Vision (ICCV)
 - European Conference on Computer Vision (ECCV)
 - International Conference on Learning Representations (ICLR)
 - Neural Information Processing Systems (NeurIPS)
 - International Conference on Machine Learning (ICML)
- Top journals
 - IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
 - International Journal of Computer Vision



Useful information

- Google scholar
 - Papers
 - Related works
 - citation
 - Authors
 - Gemini Deep Research
- Arxiv (those papers are not peer-reviewed!)
- Homepage of scholars/projects
- Github
 - Code (don't trust everything! Use simple examples to test its usability)
 - Collection of papers in certain areas (search "awesome-**")
- Youtube
 - Oral presentations of top conferences
 - Tutorials



Closing Quiz:

Purpose: Attendance + Key info check

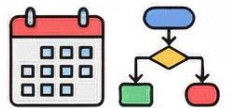
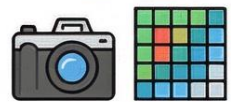
Easy questions, don't worry!



Week 1: Complete! 🎉



Anonymous
Class Feedback

- ✓ Applications 
- ✓ History 
- ✓ Course Plan 
- ✓ Image Basics 

Today's Lab:

Setup + Instagram Filters 

Next: Image Filtering 

See you next week!

